

Core Pure Mathematics 1

Notes · Edexcel A Level Further Mathematics — Core Pure 1 · Chapter 6: Matrices

Chapter 6: Matrices

Why this matters

Matrices are completely new mathematical objects at A Level, with their own rules of arithmetic that don't always behave like ordinary numbers (multiplication isn't commutative, for a start). This chapter builds the toolkit — addition, multiplication, determinants, inverses — that Chapter 7 (Linear Transformations) depends on entirely, so precision and care with the mechanics here is essential.

Concept: matrix addition and scalar multiplication

Matrices of the same size (same number of rows and columns) are added or subtracted entry by entry. A matrix is multiplied by a scalar by multiplying every entry by that scalar. Matrices of different sizes cannot be added.

Multiplication has its own, different size rule: an $(m \times n)$ matrix can only be multiplied by an $(n \times p)$ matrix — the number of *columns* in the first must match the number of *rows* in the second — and the result is an $(m \times p)$ matrix. Check this compatibility before starting any multiplication; "can I even multiply these?" is a very common early exam slip.

Key vocabulary

Term	Meaning
Order (of a matrix)	Rows $\times$ columns, e.g. a 2×3 matrix has 2 rows and 3 columns
Identity matrix, I	A square matrix with 1s on the leading diagonal and 0s elsewhere; acts like "1" for matrices
Zero matrix, O	A matrix of all zeros; acts like "0" for matrices
Determinant, $\det(M)$ or M	A single number computed from a square matrix, telling us if it has an inverse
Singular matrix	A matrix with determinant 0 (no inverse exists)
Inverse, M^{-1}	The matrix such that $MM^{-1} = M^{-1}M = I$

Determinants and inverses are only defined for **square** matrices (same number of rows and columns) — a non-square matrix can still be added, scaled or multiplied (subject to the size rules above), but it never has a determinant or an inverse.

Worked example 1: matrix multiplication

EXAMPLE 1

Question. Given $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 4 \\ 2 & 5 \end{pmatrix}$ (rows separated by semicolons), find AB .

Step 1. Recall the rule: the entry in row i , column j of AB is found by taking row i of A "dot" with column j of B . **(B1)**

Step 2. Top-left entry: $(1 \times 0) + (2 \times 2) = 0 + 4 = 4$. Top-right entry: $(1 \times 4) + (2 \times 5) = 4 + 10 = 14$. **(M1 A1)**

Step 3. Bottom-left entry: $(3 \times 0) + (-1 \times 2) = 0 - 2 = -2$. Bottom-right entry: $(3 \times 4) + (-1 \times 5) = 12 - 5 = 7$. **(M1 A1)**

Answer: $AB = \begin{pmatrix} 4 & 14 \\ -2 & 7 \end{pmatrix}$.

Exam tip

EXAM TIP

The single most important fact to learn early is that matrix multiplication is NOT commutative — $AB \neq BA$ in general. Compute BA for the same two matrices as a quick follow-up; seeing a different answer with your own hands is far more convincing than being told the rule.

Concept: determinant of a 2×2 matrix

For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is $\det(M) = ad - bc$. If $\det(M) = 0$, the matrix is singular and has no inverse; geometrically (Chapter 7), this corresponds to a transformation that squashes the plane down to a line or a point.

Concept: inverse of a 2×2 matrix

For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\det(M) \neq 0$, the inverse is:

$$M^{-1} = \frac{1}{\det(M)} \times \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

That is: swap the leading-diagonal entries, negate the other two entries, and divide everything by the determinant.

Worked example 2: determinant and inverse of a 2×2 matrix

EXAMPLE 2

Question. Find the determinant and, where it exists, the inverse of $M = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$.

Step 1. Calculate the determinant: $\det(M) = (3 \times 2) - (5 \times 1) = 6 - 5 = 1$. **(M1 A1)**

Step 2. Since $\det(M) \neq 0$, the inverse exists. Swap the diagonal entries and negate the off-diagonal entries: $\begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$. **(M1)**

Step 3. Divide by the determinant (1, so no change needed here): $M^{-1} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$. **(A1)**

Answer: $\det(M) = 1$, $M^{-1} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$.

Concept: determinant and inverse of a 3×3 matrix

For a 3×3 matrix, work in two stages:

- **Determinant:** expand along any row or column, using 2×2 "minor" determinants and alternating signs (the sign pattern $+$ $-$ $+$ $-$ $+$ $-$ $+$ is often called a checkerboard pattern).
- **Inverse** (when $\det(M) \neq 0$): find the matrix of cofactors, transpose it to get the adjugate, then divide by the determinant: $M^{-1} = (1/\det(M)) \times \text{adj}(M)$.

This is more laborious than the 2×2 case, but it's the same underlying idea — a scalar built from the entries, and a rearranged/adjusted version of the matrix divided by that scalar.

Worked example 3: determinant of a 3×3 matrix

EXAMPLE 3

Question. Find the determinant of $M = (1 \ 2 \ 3; 0 \ 1 \ 4; 5 \ 6 \ 0)$ (rows separated by semicolons).

Step 1. Expand along the top row, using the checkerboard sign pattern $(+, -, +)$: $\det(M) = 1 \times (1 \times 0 - 4 \times 6) - 2 \times (0 \times 0 - 4 \times 5) + 3 \times (0 \times 6 - 1 \times 5)$. **(M1)**

Step 2. Evaluate each 2×2 minor determinant: $(0 - 24) = -24$; $(0 - 20) = -20$; $(0 - 5) = -5$. **(M1 A1)**

Step 3. Combine with the signs and coefficients: $1 \times (-24) - 2 \times (-20) + 3 \times (-5) = -24 + 40 - 15 = 1$. **(M1 A1)**

Answer: $\det(M) = 1$.

Exam tip

EXAM TIP

For 3×3 determinants, always write out the checkerboard sign pattern above your working before starting, and cover up the row and column of the entry you're expanding by with your finger to isolate each 2×2 minor. This physical/visual habit dramatically cuts down on minor-determinant errors.

Concept: solving systems of linear equations using matrices

A system of linear equations can be written as a single matrix equation $Mx = k$, where M is the matrix of coefficients, x is the column vector of unknowns, and k is the column vector of constants. Provided M is non-singular, the unique solution is $x = M^{-1}k$.

Worked example 4: solving a linear system with a matrix inverse

EXAMPLE 4

Question. Use matrices to solve the simultaneous equations: $3x + 5y = 4$, $x + 2y = 1$.

Step 1. Write the system as a matrix equation: $(3 \ 5; 1 \ 2)(x; y) = (4; 1)$. **(M1)**

Step 2. Recognise the coefficient matrix as $M = (3 \ 5; 1 \ 2)$ from Worked example 2, with $M^{-1} = (2 \ -5; -1 \ 3)$. **(B1)**

Step 3. Multiply both sides on the left by M^{-1} : $(x; y) = (2 \ -5; -1 \ 3)(4; 1)$. **(M1)**

Step 4. Carry out the multiplication: $x = (2 \times 4) + (-5 \times 1) = 8 - 5 = 3$; $y = (-1 \times 4) + (3 \times 1) = -4 + 3 = -1$. **(M1 A1)**

Answer: $x = 3$, $y = -1$.

Exam tip

EXAM TIP

This is a great moment to connect back to GCSE/A Level simultaneous equations: check your matrix-derived answer by substituting back into both original equations, exactly as you would with elimination or substitution. It confirms the new method gives the same, correct, familiar-feeling answer.

Chapter summary

Idea	Key point
Addition/scalar multiplication	Entry-by-entry; matrices must be the same order to add
Multiplication	Row-by-column "dot products"; $AB \neq BA$ in general
2×2 determinant	$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$
2×2 inverse	Swap diagonal, negate off-diagonal, divide by determinant
3×3 determinant	Expand along a row/column with the checkerboard sign pattern and 2×2 minors
3×3 inverse	$(1/\det(M)) \times \text{adjugate matrix}$
Solving systems	Write as $Mx = k$; solve via $x = M^{-1}k$ when M is non-singular

Common exam mistakes

- Trying to multiply matrices in the wrong order, or assuming $AB = BA$.
- Trying to add/multiply matrices of incompatible orders.
- Sign errors in the 2×2 inverse formula (forgetting to negate the off-diagonal entries).
- Losing track of the checkerboard sign pattern in a 3×3 determinant expansion.
- Forgetting to check $\det(M) \neq 0$ before claiming an inverse exists.
- Multiplying M^{-1} and k in the wrong order when solving $Mx = k$ (matrix multiplication order matters).

You should now be able to

- Add, subtract and multiply matrices, including recognising when operations are undefined.
- Calculate the determinant of 2×2 and 3×3 matrices.
- Find the inverse of a non-singular 2×2 or 3×3 matrix.
- Use matrices to solve a system of linear equations.